TEAM 2025114

General Championship 2025

DATA ANALYTICS

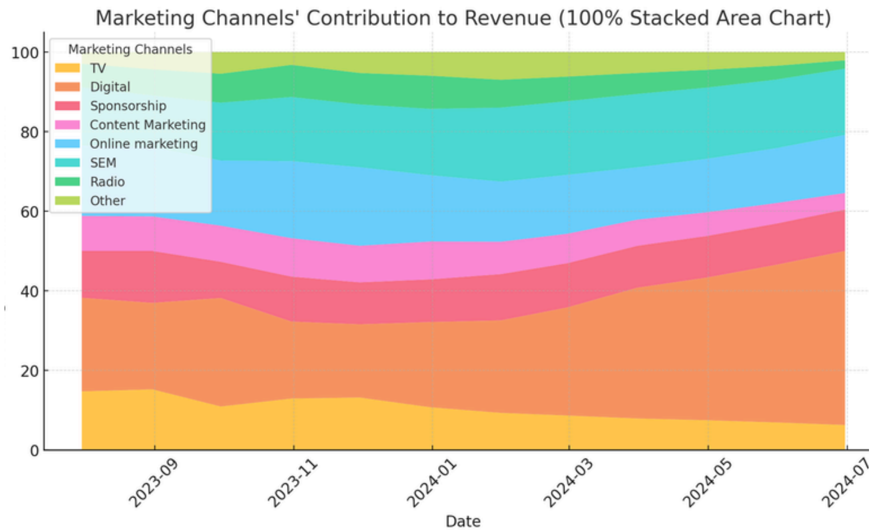
OPTIMIZING ELECTROMART'S MARKETING BUDGET WITH DATA-DRIVEN INSIGHTS

INDEX

Business Problem: Why Budget Optimization Matters? 01

ElectroMart is a Canada-based e-commerce company specializing in electronic products. Over the past year, the company has invested heavily in various marketing initiatives, including commercials, online campaigns, and large-scale promotional events. As part of its business strategy, ElectroMart aims to optimize its marketing budget for the upcoming year to maximize revenue and return on investment (ROI).

ElectroMart has been overspending on inefficient marketing channels, requiring a data-driven approach to reallocate budgets and maximize ROI.



Digital marketing channels are driving the highest ROI, while traditional media spend needs reallocation to maximize returns.
A strategic shift towards high-ROI digital platforms can optimize marketing efficiency, reduce costs, and maximize revenue impact.

Key Concern from the CFO, ElectroMart

Over the last 12 months, the marketing budget has not yielded the expected revenue impact. The CFO believes they must either cut the budget or reallocate it strategically to maximize efficiency. Marketing teams need a data-driven strategy to ensure better revenue response from future marketing investments.

Objectives

Performance driver analysis
Which KPIs drive the top-line (revenue) performance?

Impact analysis on marketing ROI
What is the quantitative impact of each commercial lever on revenue?

Optimizing marketing spending
Best allocating the marketing budget for highest outcome

Data overview & Methodology 02

Marketing Mix Modeling (MMM)

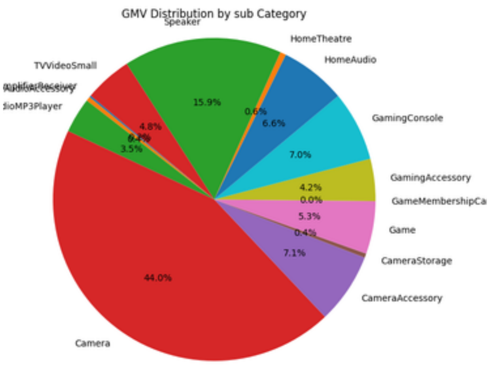
Marketing Mix Modeling (MMM) is a statistical approach used to quantify the impact of various marketing channels on sales while accounting for external factors like seasonality, economic conditions, and competitor activity. It helps identify high-ROI channels and optimize budget allocation for maximum efficiency.

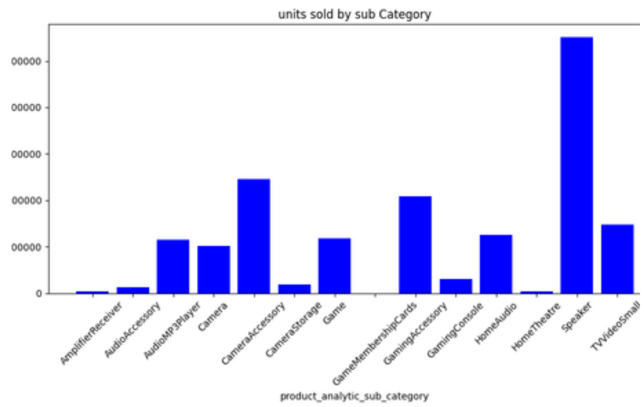
Used Datasets

For accurate marketing budget optimization, we utilized multiple datasets containing historical marketing performance, revenue, and external factors. Below are the datasets used in the analysis:

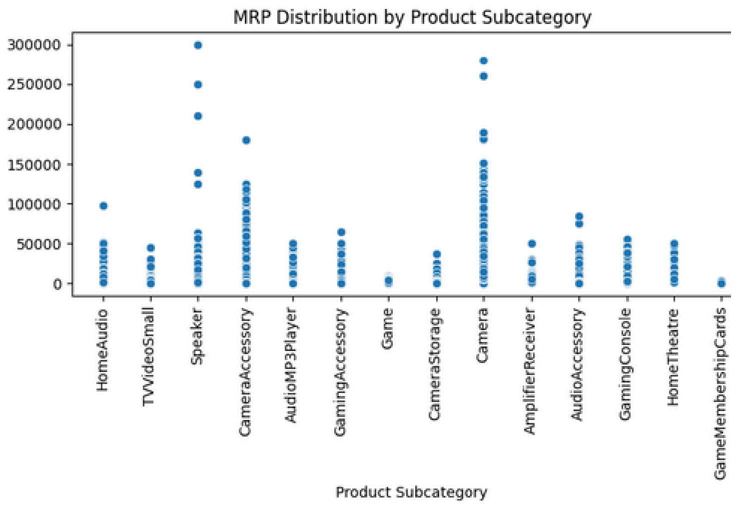
Dataset Name	Description	Key Variables Used
Customer Orders Data	Contains details of past purchases, revenue, and product categories.	Order ID, Revenue, Product Type, Discount %
Media Spend Data	Marketing budget allocation across different channels.	TV Spend, Digital Ads Spend, Social Media Spend, Print Media, Sponsorships
Sales Calendar Data	Promotional events, holidays, and seasonal effects.	Black Friday, Cyber Monday, Holiday Sales
NPS & Customer Satisfaction	Net Promoter Score & stock index for brand perception analysis.	NPS Score, Customer Retention, Stock Index
Weather Data	Regional climate trends that impact sales.	Temperature, Rainfall, Snowfall
Holiday Calendar Data	National & festive holidays affecting sales trends.	Public Holidays, Payday Effect
Product Hierarchy Data	SKU details and category classification.	Super-category, Sub-category, SKU Mapping

Exploratory Data Analysis 03





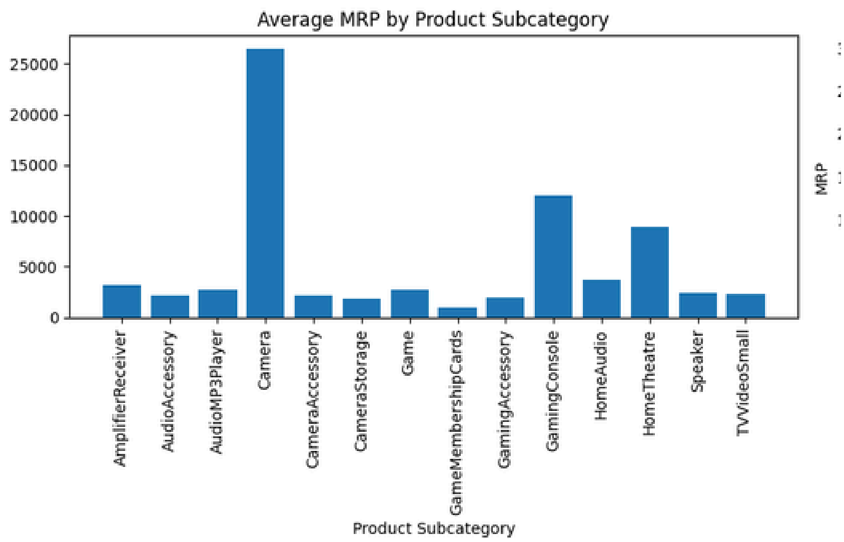
Cameras dominate GMV (44%), indicating they are high-value, low-volume products, whereas Speakers lead in sales volume but generate lower revenue. Implement premium pricing and exclusive marketing for Cameras to maximize profit, while leveraging bulk discounts and strategic bundling for high-sales categories like Speakers to drive overall revenue growth.



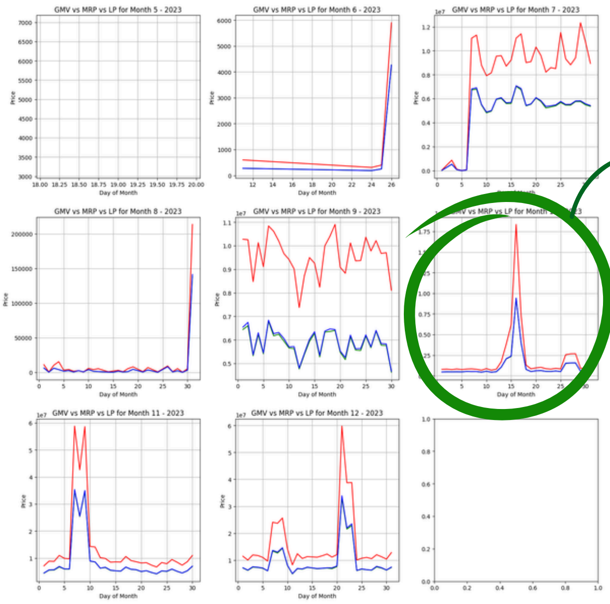
Cameras, Speakers, and Camera Accessories have a wide price range, catering to both budget and premium customers—ElectroMart should segment its marketing strategy, promoting premium models through high-end branding and financing options while using discounts and bundles to push budget-friendly products.

Recommendation

To maximize profits, the company should emphasize premium pricing and exclusive offers on high-MRP products while driving volume sales for lower-priced variants through targeted promotions, festive discounts, and combo deals.



GMV vs MRP vs LP month wise for 2023



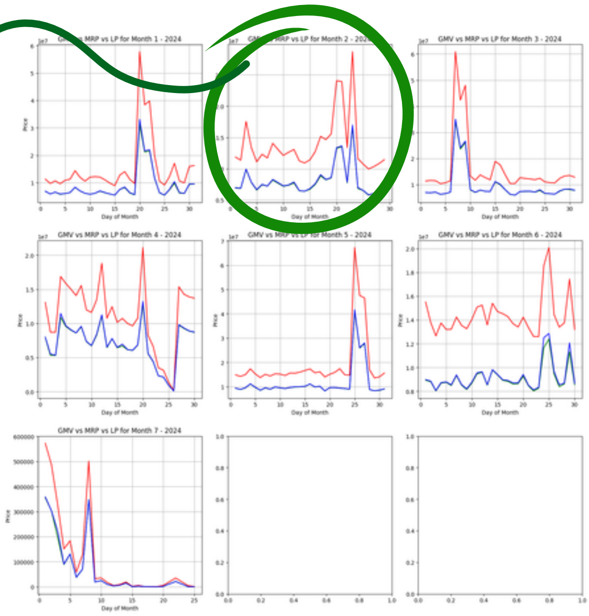
Best Month - October 2023

Mid-Month Sales Peak: A sharp spike in GMV, MRP, and LP suggests a major promotional event driving high sales.
Discount-Driven Growth: A widened gap between MRP and LP indicates heavy discounting played a key role in boosting revenue.
Post-Sale Decline: Sales dropped after the peak, highlighting the need for strategies to maintain demand beyond promotions.

GMV vs MRP vs LP month wise for 2024

Best Month - February 2024

Major Sales Surge: A sharp peak in GMV, MRP, and LP around mid-to-late February indicates a successful promotional event.
High Discount Impact: The large gap between MRP and LP suggests that deep discounts played a crucial role in driving sales.
Post-Promotion Drop: Sales decline after the peak, highlighting the need for sustained marketing efforts to maintain demand beyond sales periods.



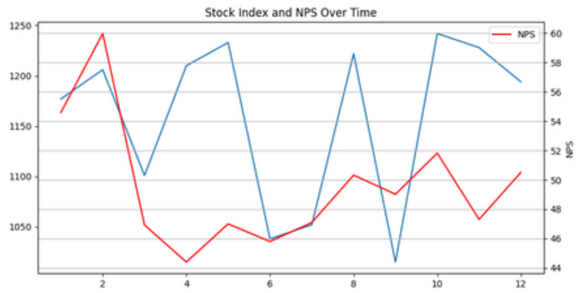
NPS vs Stock Index

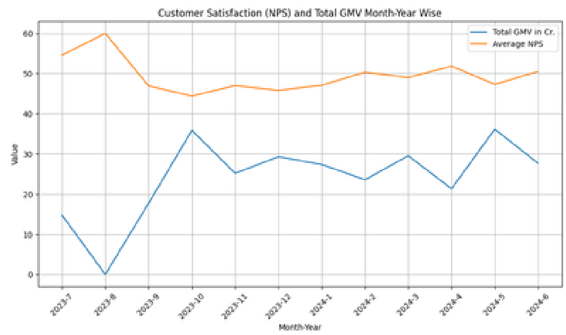
Divergence

NPS and the stock index sometimes diverge, indicating that shifts in customer sentiment don't always align with market performance, as both can be influenced by different factors.

Potential Correlation

At times, both metrics move together, suggesting a potential link where higher customer satisfaction may drive better market performance, or market success could boost customer sentiment.

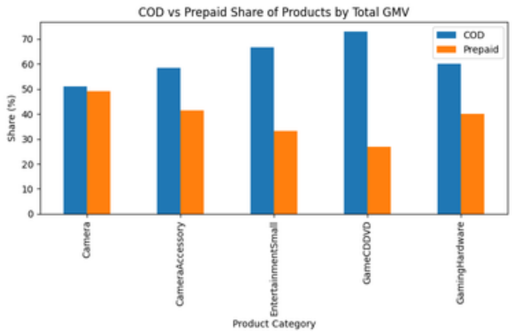


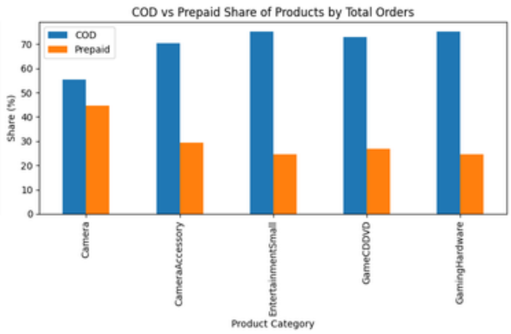


NPS vs Total GMV Month-Year wise

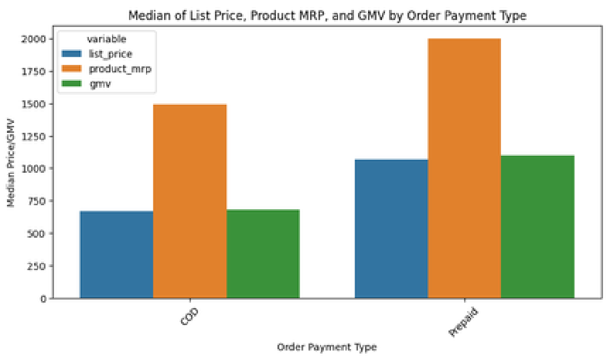
NPS Stability vs. GMV Fluctuations: NPS remains stable despite GMV variations, indicating customer loyalty isn't impacted by short-term revenue dips.
Lag Effect in Customer Impact: NPS growth often precedes GMV growth, suggesting a delayed impact of customer satisfaction on revenue.
Leveraging High-NPS Customers: Engage satisfied customers with exclusive offers during slow periods to convert loyalty into repeat purchases.

COD vs Prepaid



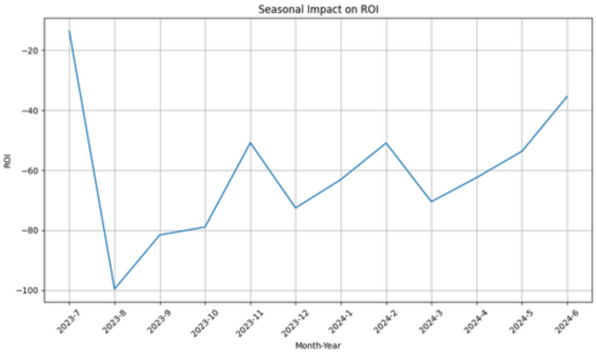


COD Dominates Sales but Prepaid Brings Higher Value: COD holds 70%+ of GMV and orders, but prepaid customers spend more per order, making prepaid adoption essential for long-term growth.
Encourage Prepaid with Incentives: Offer cashback, faster delivery, and EMI options to shift COD customers toward prepaid, improving cash flow and reducing return risks.



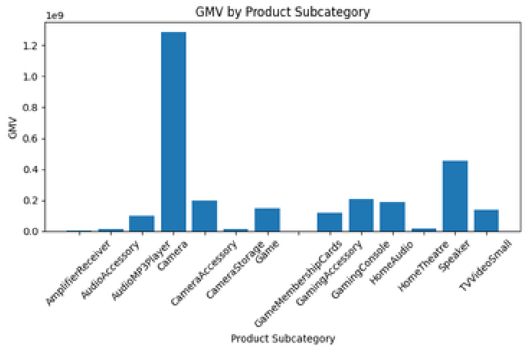
Cameras and Accessories already attract prepaid buyers—offering better financing, EMI options, and premium positioning can further boost conversions.
Use COD's popularity to increase order value through bundling, cross-selling, and targeted loyalty programs, gradually shifting customers toward prepaid transactions.

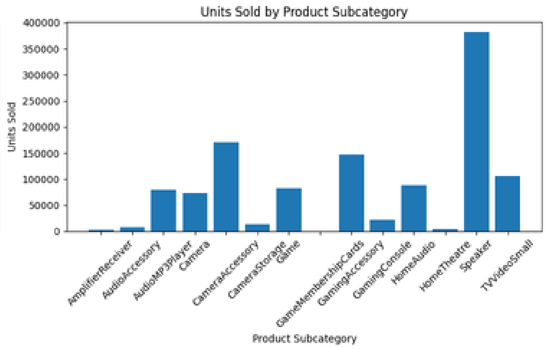
Seasonal Impact on ROI



Fluctuating ROI: Significant dips, especially mid-2023, suggest seasonal impacts.
Strategic Adjustments: Reduce costs during low ROI periods and boost sales with targeted promotions.
Resource Allocation: Maximize investment in high-ROI months and use off-peak times for planning and optimization.

Product Performance & Strategic Recommendations

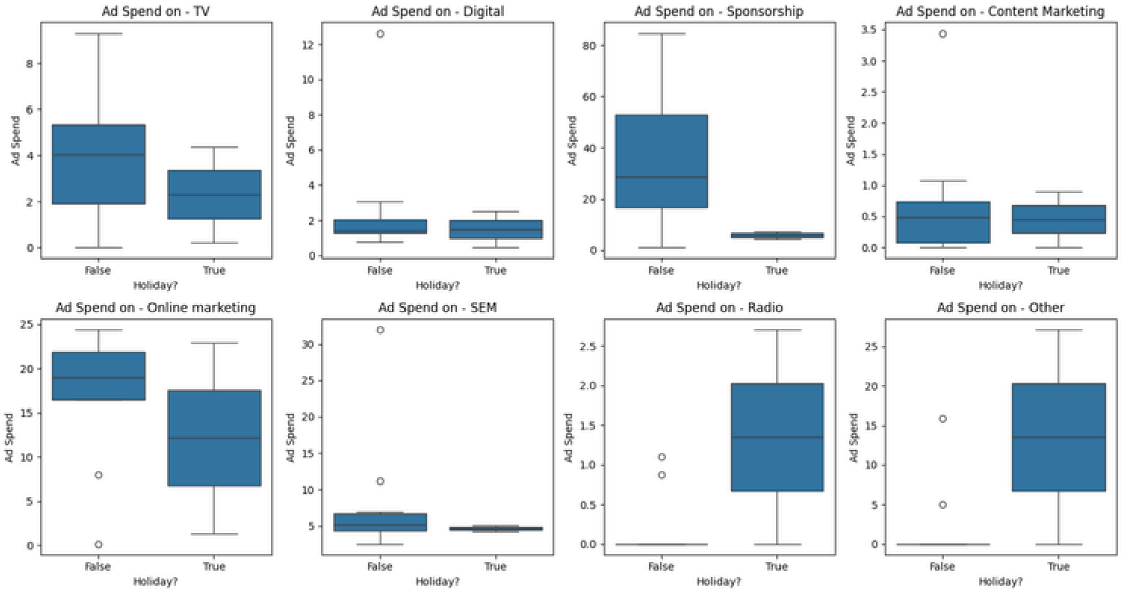




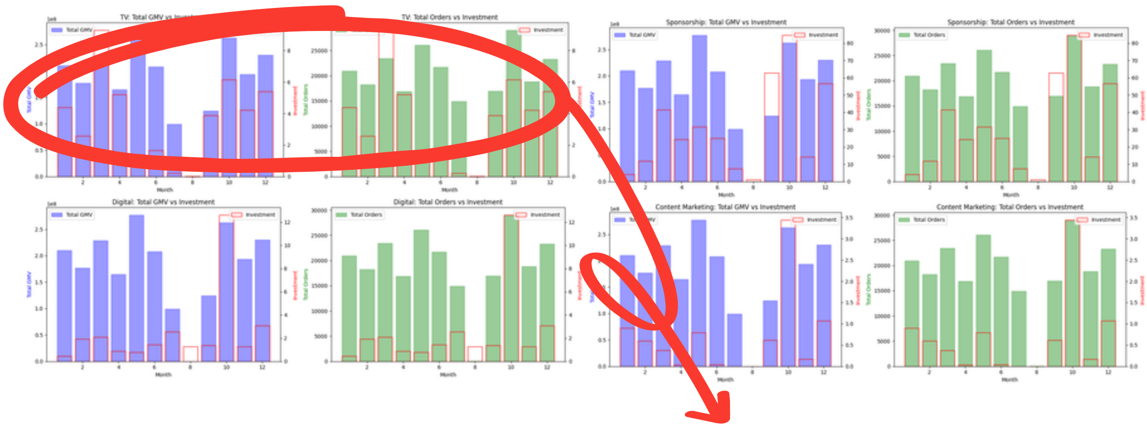
1. **Cameras (44% GMV) and Gaming Consoles (7% GMV) have high value but moderate sales, requiring premium positioning, financing, and trade-in offers.**
2. **Speakers lead in volume (15.9% GMV) but need upselling and bundling to improve margins. Gaming Accessories & Membership Cards should leverage subscriptions and console bundles for better profitability.**
3. **Underperforming categories like Amplifiers and Camera Storage should be repositioned or phased out, while Camera and Gaming Accessories can grow through bundled promotions and influencer marketing.**

Holidays vs Non-Holidays

TV & Sponsorship see higher spend outside holidays, focusing on brand awareness, while Online & Other channels increase during holidays for seasonal promotions. Digital & Content Marketing remain steady year-round, ensuring consistent ROI. Radio sees slight holiday spikes for mass reach. Focus TV & Sponsorship on brand-building outside holidays while boosting digital ads during peak seasons for higher conversions. Maintain consistent investment in content marketing for long-term customer engagement. Optimize budget by prioritizing high-ROI channels (TV, Digital, Affiliates) and reducing low-impact spending.

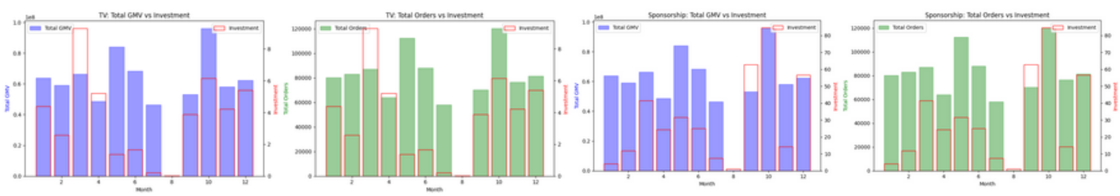


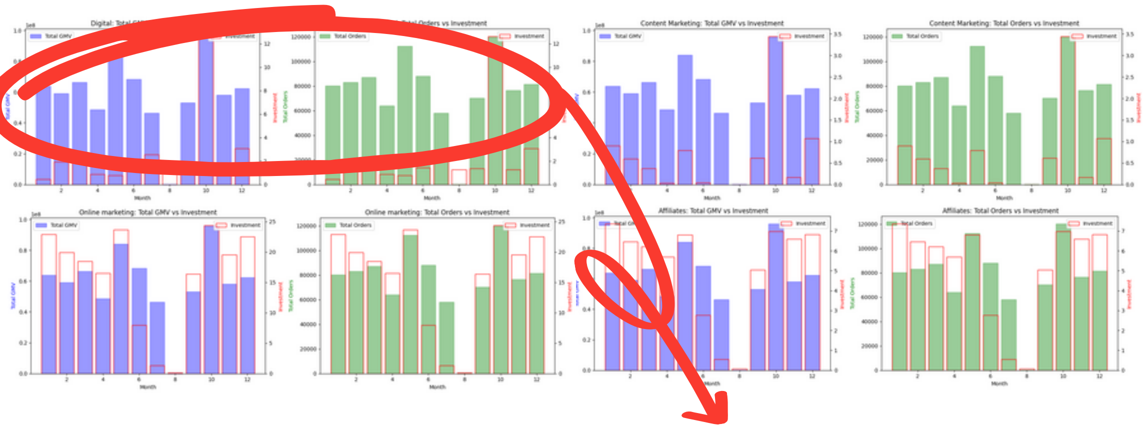
Total GMV vs Investment and Total GMV vs Orders for Luxury products



The analysis reveals that TV marketing investment does not consistently translate into higher GMV or order volume for luxury products, highlighting inefficiencies in spend allocation. While some months see a strong correlation between investment and returns, others (e.g., month 8) show disproportionate spending with minimal impact, signaling potential wastage. To maximize ROI, a strategic shift toward high-performing digital and premium-targeted marketing channels is crucial for luxury product success.

Total GMV vs Investment & Total GMV vs Orders for Mass Market products

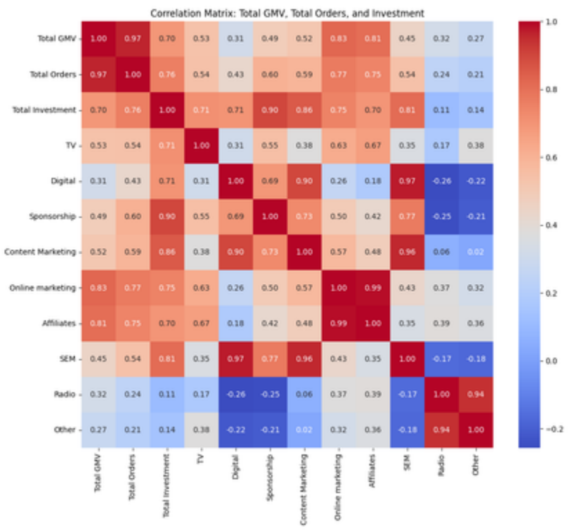




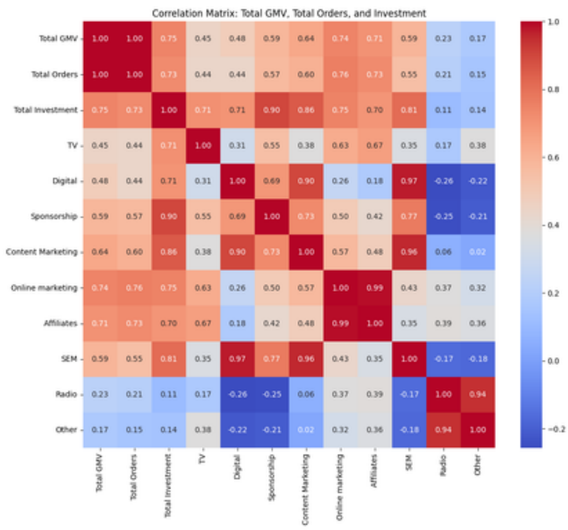
The analysis shows that digital marketing investment strongly correlates with increased GMV and order volume for mass-market products, making it a high-impact channel. Unlike TV advertising, digital spend delivers consistent returns, especially in peak months like month 10, where a spike in investment directly drives record-high sales. To sustain and amplify growth, scaling digital investment strategically while optimizing spend efficiency can maximize revenue impact in the mass-market segment.

Correlation matrix: Total GMV, Total Orders, and Investment

Luxury products



Mass Market products



Online Marketing (0.83), Affiliates (0.81), and Total Investment (0.70+) drive high GMV, while TV (0.53) and Digital (0.31) have a weaker impact.

Total Investment (0.75), Online Marketing (0.74), and Content Marketing (0.64) strongly influence GMV, with Digital (0.48) performing better than in luxury.

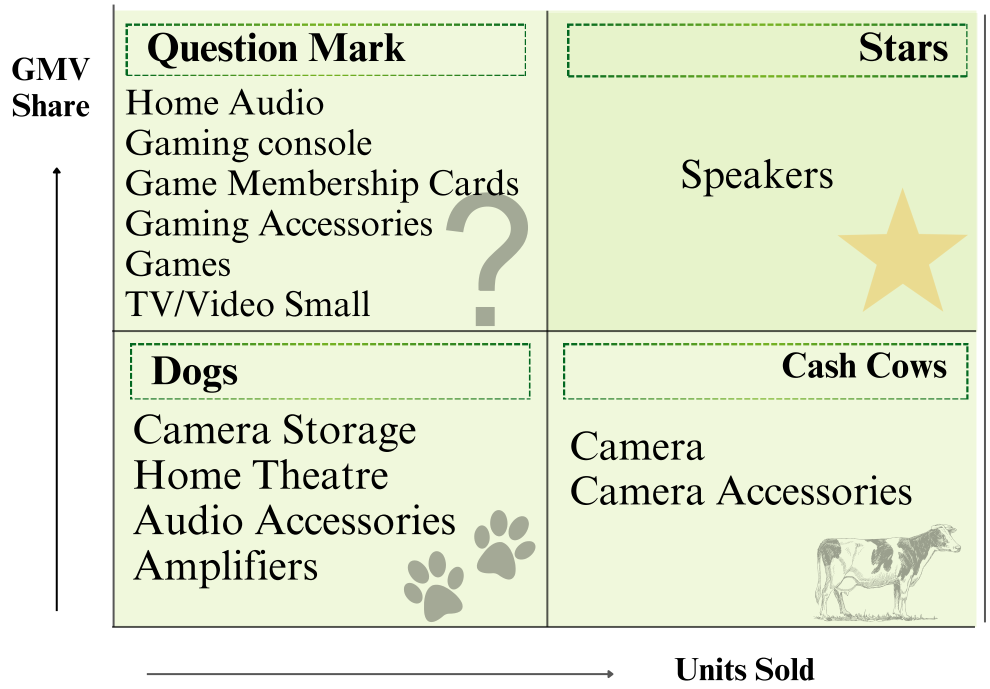
Key Performance Indicators (KPIs) 04

KPIs	Strategy	Benchmark	Actionable Recommend-ations	Expected Business Impact	Potential Risks	Implementat-ion Time
Correlation b/w sales & marketing spends	**Prioritize high-ROI channels (Online, Affiliates)** and optimize TV & Digital spend.	Target ROI increase of 15% by shifting 20% budget to high-performing channels.	Reallocate budget towards digital and online channels while limiting offline inefficiencies.	Increased sales efficiency and improved ROI by 15-20%	Fluctuations in digital ad performance may impact marketing ROI	Short-term (0-3 months): Adjust budget allocation for high-ROI channels.
Product Positioning Matrix	**Invest more in 'Stars' for growth,** maintain 'Cash Cows' for profitability, and optimize 'Question Marks'	Increase revenue share of 'Stars' by 20% and reduce 'Dogs' share by 10%	Develop promotional strategies for 'Stars' and phase out 'Dogs' with low sales impact	Higher revenue from well-performing products and reduced losses from low-selling ones.	Improper categorization may lead to misallocation of resources.	Short-term (0-3 months): Realign investment strategies based on BCG matrix.
Top 3 Categories (GMV & Units Sold Trend)	**Increase investment in the top 3** performing categories to drive revenue.	Achieve 15% YoY growth in top-performing categories.	Introduce bundle offers and targeted discounts in top-selling categories.	Boosted revenue by leveraging demand in strong-performing categories.	Demand shifts could impact performance of top categories	Mid-term (3-6 months): Optimize marketing for top-performing categories
Luxury vs. Mass Market	Luxury relies more on **Online & Affiliates**; Mass Market benefits from **Digital & Content Marketing**.	Optimize spend allocation to increase high-margin luxury sales by 10%.	Customize luxury campaigns for high-net-worth customers, focusing on digital and affiliate channels	Improved luxury product profitability and enhanced high-value customer retention	Market trends and customer behavior may vary, requiring continuous analysis.	Mid-term (3-6 months): Implement luxury-focused digital ad campaigns

KPIs	Strategy	Benchmark	Actionable Recommend-ations	Expected Business Impact	Potential Risks	Implementa t-ion Time
Customer Acquisition Cost (CAC)	**Optimize acquisition costs** by balancing paid marketing with organic strategies	Maintain CAC below 10% of Average Order Value (AOV)	Balance digital marketing spend with organic lead generation strategies.	Lower acquisition costs and better long-term profitabilit y	Unpredicta ble cost fluctuation s may affect acquisition expenses	Mid-term (3-6 months): Introduce CAC monitoring and optimizatio n strategies
Product MRP Distribution	Adjust pricing strategy based on MRP distribution to enhance revenue and market reach.	Ensure MRP alignment with competitor pricing for better market positioning.	Adjust pricing dynamically based on competitor analysis and demand forecasting.	Higher customer conversion rates due to competitive pricing strategies.	Pricing misalignme nt with competitor s may reduce conversions .	Short-term (0-3 months): Align pricing with market trends.
COD vs Prepaid GMV Share	Encourage prepaid payments with discounts and reduce COD-related costs.	Increase prepaid transactions by 10% to improve cash flow and reduce logistics costs.	Offer exclusive benefits for prepaid transactions to encourage adoption.	Reduced COD risks and improved cash flow manageme nt.	Customer preference for COD may persist despite incentives.	Short-term (0-3 months): Implement prepaid discount strategies.
Correlation of Units Sold Across Products	Optimize inventory and pricing by understandin g demand correlations among products.	Identify at least 3 strong product correlations for better inventory planning.	Use correlation insights for inventory optimization and targeted marketing.	Better stock manageme nt and improved marketing efficiency.	Seasonality and external factors may affect correlation patterns.	Short-term (0-3 months): Improve stock planning using correlation analysis.
Bivariate Analysis: Sales vs. Orders & Spending	Align marketing spend with sales trends to ensure effective budget allocation.	Ensure a minimum 0.7 correlation between marketing spend and sales trends.	Reallocate marketing spend based on high-conversion months and seasonal demand.	Optimized marketing investments with better alignment to sales performanc e.	Changes in consumer behavior could alter previous trends.	Mid-term (3-6 months): Refine marketing spending patterns.

KPIs	Strategy	Benchmark	Actionable Recommend-ations	Expected Business Impact	Potential Risks	Implement at-ion Time
Ad Spend by Month	Shift ad spend to high-performing months and reduce inefficient spending periods.	Improve ad spend efficiency by cutting underperforming months by 15%.	Monitor seasonal trends and optimize budget allocation based on past ad performance.	Reduced ad waste and increased returns on marketing investment.	Unforeseen market trends may require flexible ad strategies.	Short-term (0-3 months): Adjust ad spend based on historical trends.
Ad Spend on Holiday vs. Non-Holiday	Increase holiday-specific promotions for better seasonal conversions.	Increase holiday-season GMV by 20% through optimized marketing campaigns.	Launch aggressive promotions during peak shopping seasons to maximize conversion rates.	Better holiday season sales and improved customer targeting.	Holiday promotions may face high competition from other retailers.	Mid-term (3-6 months): Execute holiday-focused campaigns.
Sales in Different Temperature Ranges	Target warm-weather sales for high-performing products and plan discounts for snowfall periods.	Align sales peaks with optimal weather conditions to maximize revenue.	Develop targeted pricing strategies for different weather conditions.	Higher sales in optimal weather conditions and seasonal adjustments.	Weather unpredictability could affect forecasting accuracy.	Long-term (6+ months): Implement weather-based demand strategies.
NPS Score	Improve customer experience through targeted engagement strategies based on NPS insights.	Increase NPS score to above 50 for strong customer satisfaction.	Implement customer feedback systems and loyalty programs to improve satisfaction.	Stronger customer loyalty and higher retention rates.	Negative feedback loops could impact brand reputation.	Long-term (6+ months): Establish continuous customer satisfaction improvements.

Product Positioning Matrix 05



This categorizes ElectroMart's products based on GMV share (revenue contribution) and units sold (market share). It adapts the BCG Matrix for internal analysis, helping optimize budget allocation. GMV Share highlights revenue drivers, while units sold shows product popularity.

Impact Analysis on Marketing ROI 06

1 Sales Peaks During Discounts

Discounts boost GMV by increasing customer traffic, leading to higher sales volumes that outweigh margin reductions, ultimately driving overall revenue growth.

2 Holidays vs. Paydays on Sales

Holidays boost sales, while paydays have minimal impact, making holiday promotions more effective.

3 High ROI Channels

Content Marketing has the highest ROI, while Digital & TV ads also deliver strong revenue returns.

4 Budget Allocation

Sponsorships consume 43.17% of the budget but show low ROI, while Online Marketing (22.88%) has moderate efficiency, needing optimization.

Marketing Channel	Spend (₹ Cr)	ROI (ratio)	CAC (₹)	Budget Action
TV	44.368068	5.501802	468.104160	Maintain Budget
Digital	29.660227	8.725893	312.929694	Increase Budget
Sponsorship	365.367769	-0.210461	3854.806047	Reallocate/Eliminate
Content Marketing	8.027973	34.933377	84.698989	Increase Budget
Online Marketing	193.639614	0.489737	2042.991247	Reallocate/Eliminate
Affiliates	61.36784	3.700557	647.480801	Reduce Budget
SEM	91.202143	2.162998	962.226565	Reduce Budget
Radio	4.670000	60.771348	49.270751	Increase Budget
Other	48.022545	5.007016	506.660995	Maintain Budget

Statistical Analysis 07

Models	R2 Score	MSE	Insights
Decision Tree	0.9962	102084.9613	It has highest R2 Score and lowest error
AdaBoost	0.8743	3883002.955	Low R2 score suggests that it might struggle with capturing relationships
Linear Regression	0.8486	4074827.72	Similar to Multiple Linear Regression, providing a solid baseline
Lasso Regression	0.8942	2848980.57	Improves generalization by reducing overfitting, making it more stable for moderately complex datasets.
XGBoost	0.9825	470731.803	High R2 score, has both high accuracy and stability on large data

BEST MODEL : Decision Tree

R^2 score is 0.966, indicating an excellent fit to the data, and one of the lowest Mean Squared Errors (MSE = 1093852.96), suggesting that its predictions are highly accurate.

Budget & Strategy

Promotional Strategy

Promotion Type	Impact	Strategy
Holiday Promotions	Significant increase in GMV and customer engagement	Increase holiday season budget for digital ads and online campaigns
Discount-Driven Sales	Discounted products see spikes in sales, proving their effectiveness	Continue seasonal discount promotions with strategic stock planning
Ad Spend on Digital & TV	Digital Ads show immediate impact, TV Ads contribute to long-term brand awareness	Allocate more towards Digital for immediate conversions and TV for awareness
Radio & Content Marketing	High ROI but low budget allocation	Increase investment moderately to test further impact
Sponsorship Budget Reallocation	Sponsorships consume too much budget but don't yield proportional sales impact	Reduce sponsorship spending and focus on direct digital conversion channels

Product Specific Recommendations

Product Category	Marketing Channel Effectiveness	Strategic Recommendation
Luxury Products(High margin, Low Volume)	Digital Ads, Content Marketing, and Influencer Sponsorships work best	Focus on premium branding, influencer campaigns, and financing options
Mass-Market Products (Low margin, High volume)	Online Ads & Discount Campaigns drive higher sales	Implement aggressive discount campaigns and performance marketing strategies
Luxury Market Growth	Customers show interest but require more engagement	Increase Search Engine Marketing (SEM) and personalized promotions
High-Volume, Low-Margin Sales	Heavy dependence on discount seasons	Increase year-round promotions and optimize conversion rates

Promotional Strategy

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Budget Optimization Strategy

Ad Channels	Budget Allocated (Rs)
TV	147822.137117
Digital	442495.557817
Sponsorship	400099.431852
Content Marketing	43039.124251
Online Marketing	183469.001652
Affiliates	24561.326797
SEM	837842.668157
Radio	899999.999999
Other	3096.562743

Maximum GMV (estimated) per order = Rs 5993.40252036595

Monthly revenue generated = 52.1689 Cr

ANNEXURE

Data Preprocessing

Data cleaning and handling missing Values
Datasets used:

1. **Stock index and NPS score: made the first row as column names to improve usability.**
2. **Canada holiday list: extracted week no. for each holiday**
3. **Media data file: filled null values with 0**
4. **Weather datasets: deleted columns with no numeric values and rows with missing values were filled by linear interpolation.**

Further processing:

1. **Created a new dataframe with gmv sum for each month**
2. **Processed customer order data to calculate monthly GMV (Gross Merchandise Value), merged it with investment and stock data, and computed Return on Investment (ROI) for each month.**
3. **Extraction of year and month from customer orders data**

Feature Engineering:

1. **Luxury and mass market product bifurcation: If the GMV Values lie above 80 percentile, theproduct is considered to be a Luxury product, else mass market product.**
2. **Customer Acquisition cost: Here we find the new customers every month and total money spent on marketing every month.**
3. **CAC is calculated by Total spend/No. of New Customers**

Exploratory Data Analysis

Discount vs Sales: Find the discount for every order and found the data for GMV of each month.
Customer Satisfaction (NPS) and Sales: Plotted customer satisfaction(NPS) and total gmv (yyyy-mm) wise.
COD vs Prepaid share of products: Analysed share of COD and prepaid payments based on total GMV and Total Orders, median of order value, List Price, Product MRP and GMV.
Seasonal impact on sales: analyzed seasonal impact on ROI over different months. Here ROI means GMV produced as return on marketing investments.
GMV Share and units sold of each sub category of products: helped us buold the product positioning matrix
MRP Distribution for each sub category: analysed average MRP and price ranges of each sub category.
Top 3 Categories: analysed top 3 categories based on GMV and Units sold in each month

Exploratory Data Analysis (Continued)

- **Monthly spending on each advertisement channels: It tells us how much money was invested on each advertisement channel in each month.**
- **Ad expense on holidays and non-holidays: It tells how much money was spend on holidays and non-holidays**

Bivariate data analysis:

- **Separating Luxury and Mass Market data and merged it with Investment & Stock Data: Merging of luxury and mass market GMV data with investment and stock data**
- **Creating twin bar charts: Two y-axes (GMV/orders and Investment)**

1. **Plot 1: Total GMV vs Month and Investment vs Month**
2. **Plot 2: Total orders vs Month and Investment vs Month**
3. **Colour code: blue - Total GMV, green - Total orders, red - Investment**

- **Plotted twin graph for selected marketing channel for luxury and mass market products**
- **Correlation analysis: Correlation between Total GMV, Total Orders, Total Investment and marketing spends for Luxury products and Mass Markets**
- **ROI analysis based on investment: Tells the Return on Investment month wise, for each advertisement channel.**

Model Implementation

Data Integration & Feature Engineering:

- **Combined datasets (orders, marketing spend, weather, holidays) and engineered features like product type, discount %, holiday weeks, and one-hot encoded categories.**

Model Training & Evaluation:

- **Trained multiple models (Linear Regression, XGBoost, LightGBM, etc.) to predict monthly GMV using marketing and product-related features. Evaluated using R^2 and MSE scores.**

ROI & Feature Analysis:

- **Used Random Forest to identify top-performing marketing channels. Channels like Content Marketing, TV, and Digital showed high ROI.**

Non-linear Response Modeling:

- **Plotted sales vs. spend curves for each channel using polynomial regression to visualize diminishing returns and optimal investment ranges.**

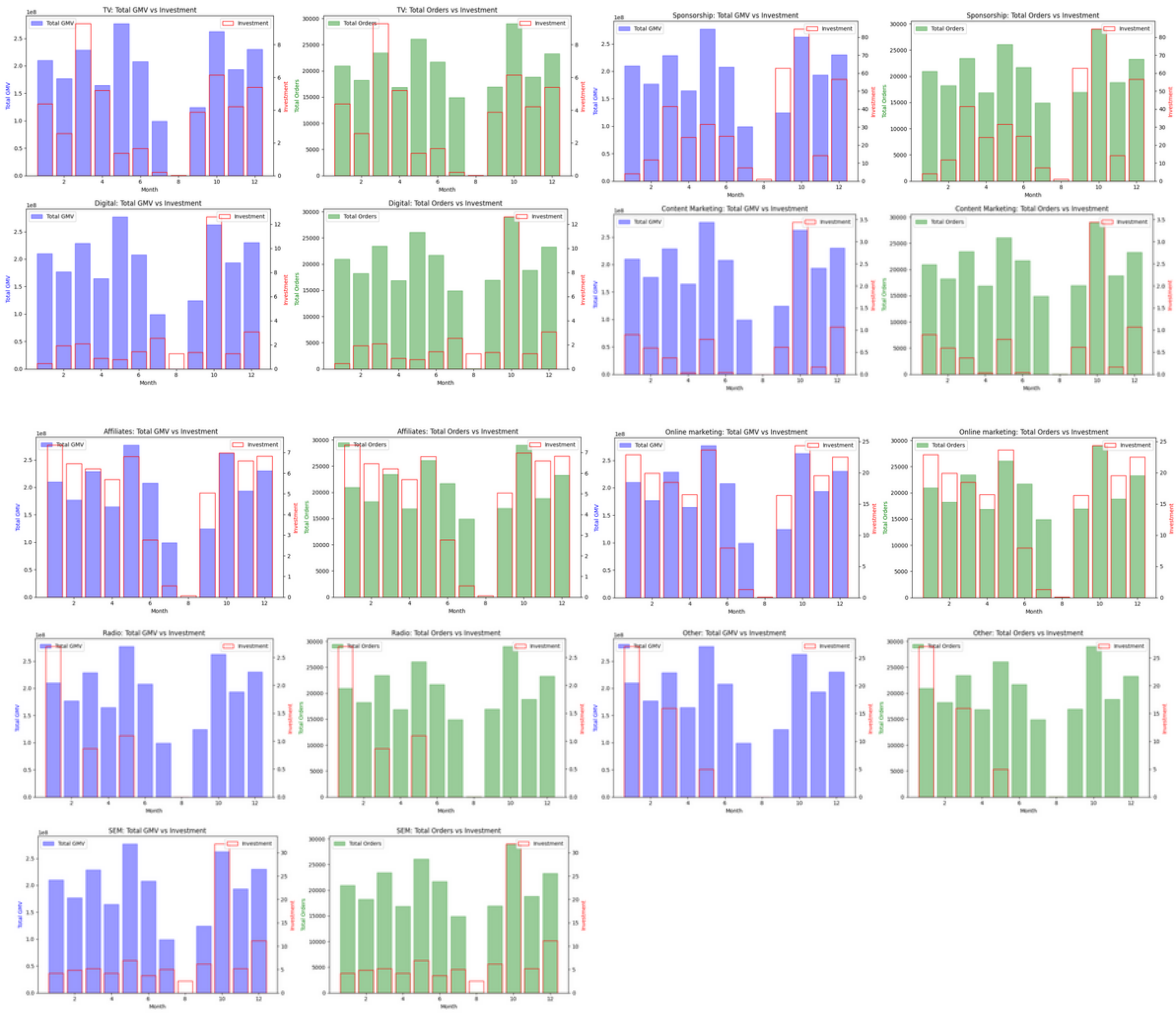
Budget Optimization Model:

- **Built an optimization function (based on Linear Regression) to maximize GMV under total budget constraints using SLSQP algorithm from scipy, optimize.**

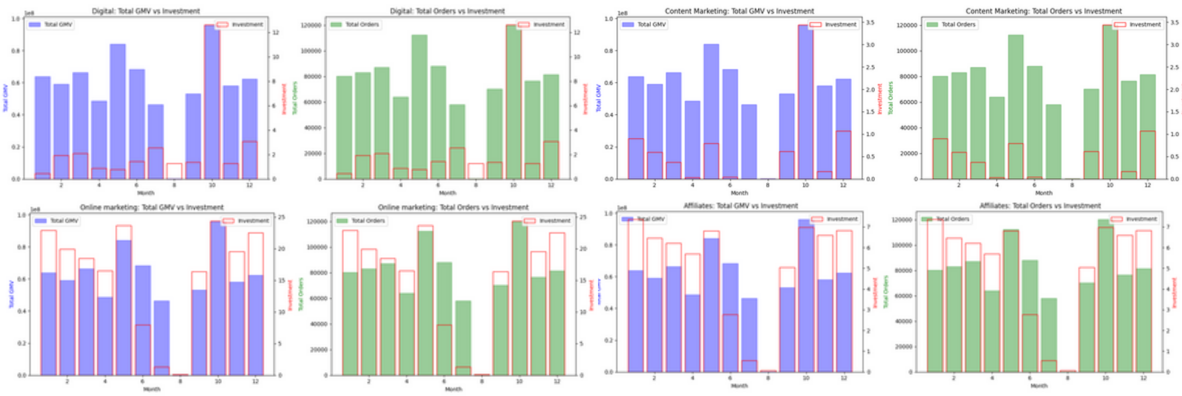
Final Outcome:

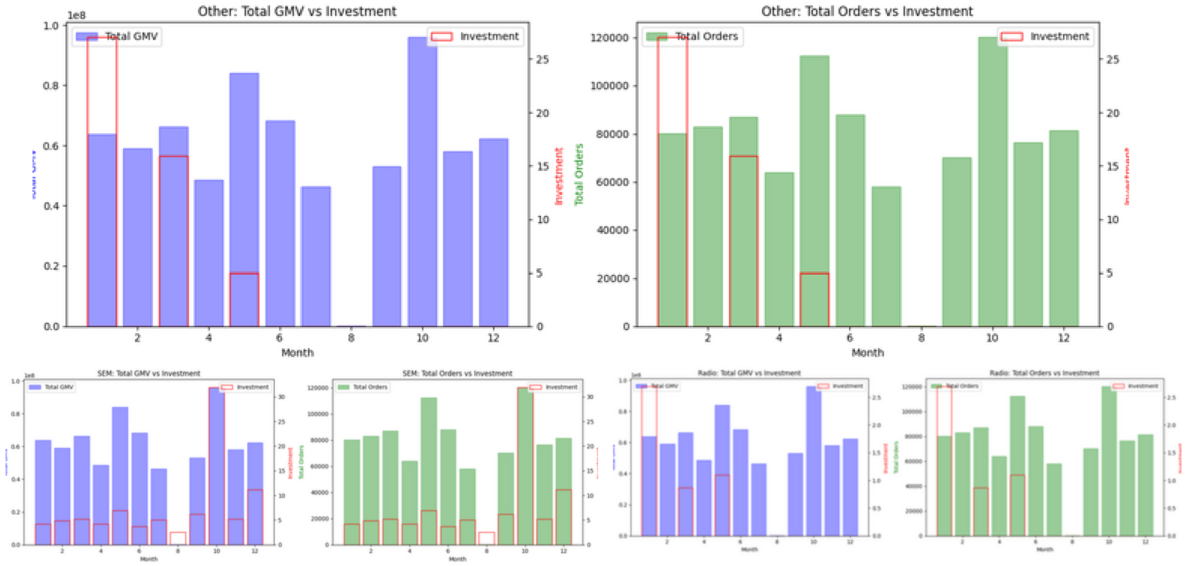
- **The optimized budget allocation is projected to boost GMV to ₹5,993.40, redistributing investments towards high-impact channels like Radio, Digital, and Content Marketing.**

Total GMV vs Investment & Total GMV vs Orders for Luxury products

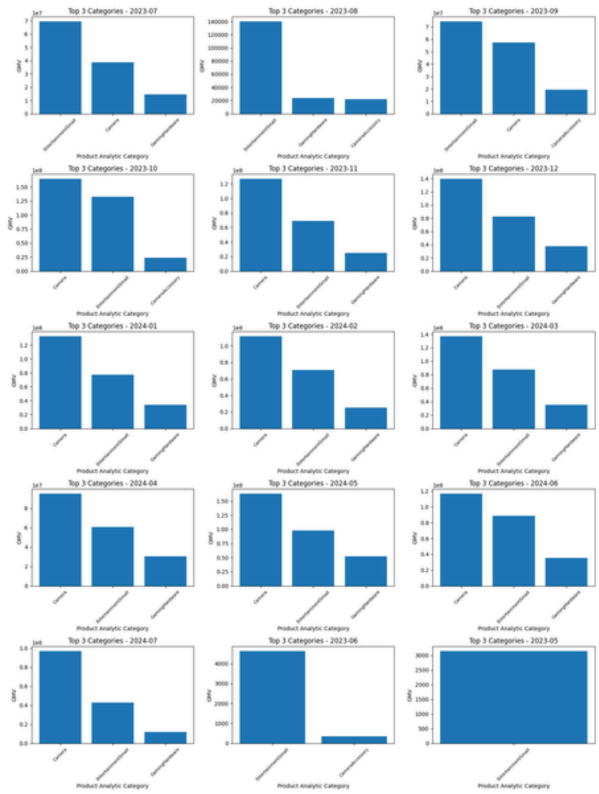


Total GMV vs Investment and Total GMV vs Orders for Mass Market





Top 3 categories for GMV



Top 3 categories for Orders